

Some Introductory Concepts [8.1-8.3]

Objective: Review of basic thermodynamic concepts (including 2nd law); develop isentropic equations

Brief Review of Thermodynamics [8.1]

p : pressure
 ρ : density
 T : absolute temperature

$$p = \rho RT$$

$\left(R = \frac{\bar{R}}{M} \rightarrow \begin{array}{l} \text{universal gas constant (1.10)} \\ \text{mass of one mole} \end{array} \right)$
 $\bar{R} = 8314 \text{ J/kg}^\circ\text{K}$
 $M = 28.97 \text{ kg/mole}$

$$R = \text{specific gas constant} = 287 \text{ J/kg}^\circ\text{K} = 1716 \text{ ft} \cdot \text{lb}/(\text{slug} \cdot \text{R})$$

Or $p\nu = RT$ ν specific volume = $\frac{1}{\rho}$

this is accurate (< 1% error) for low pressures ($\leq 1 \text{ atm}$)

moderate temperatures ($\geq 273\text{K}$)

Specific Heats [8.1.1-2]

internal energy: $u_e \equiv u_e(v, T)$ (8.1)

enthalpy: $h \equiv h(T, p) = e + pv$ (8.6)

thermally perfect gas: $u_e = u_e(T), h = h(T)$

$$du_e = c_v dT, \quad dh = c_p dT \quad (8.3, 8.10)$$

for c_p, c_v constant $\Rightarrow$ calorically perfect gas

$$u_e = c_v T \quad \text{where:} \quad c_v = \left(\frac{\partial u_e}{\partial T} \right)_v \quad (8.3)$$

$$h = c_p T \quad \text{where:} \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p \quad (8.11)$$

$$c_p - c_v = R \quad \gamma \equiv \frac{c_p}{c_v} \quad (8.15)$$

$$\gamma = \frac{n+2}{n}$$

n = 3
 = 1.29 for monatomic gases
 = 1.4 for diatomic gases *≈ air*
n = 5

n = # of molecular degrees of freedom

= 1.67 for triatomic gases, *eg. CO₂*
n = 3

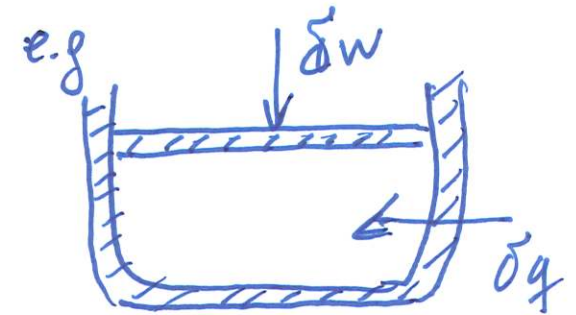
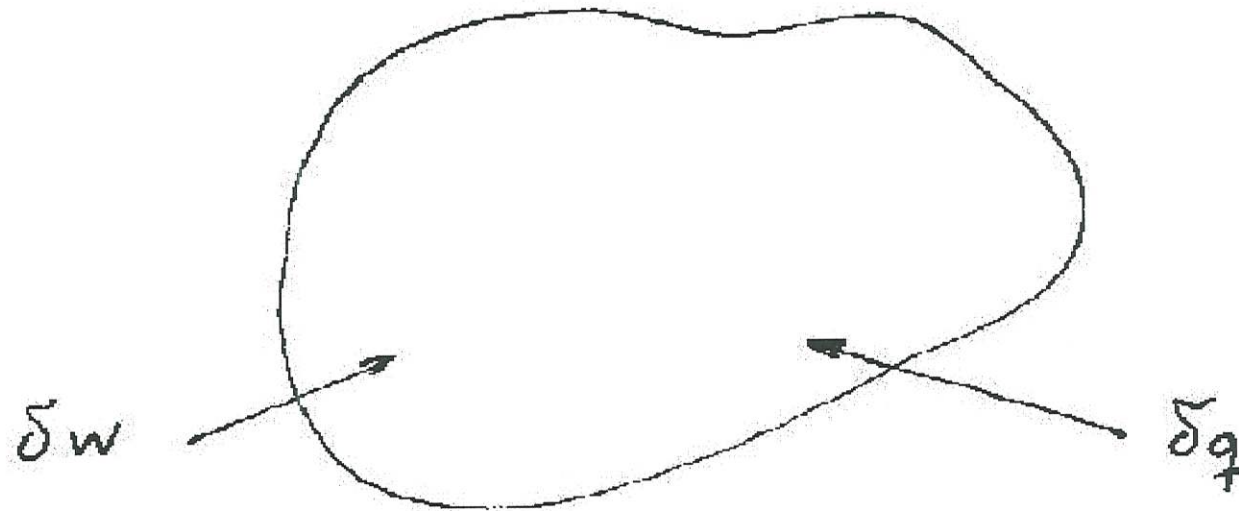
$$\Rightarrow 1 - \frac{c_v}{c_p} = \frac{R}{c_p} \Rightarrow 1 - \frac{1}{\gamma} = \frac{R}{c_p} \Rightarrow$$

$$\Rightarrow c_p = \frac{\gamma R}{\gamma - 1} \quad \text{and} \quad c_v = \frac{R}{\gamma - 1} \quad (8.16)$$

can be found for air (R, γ are known)

First Law of Thermodynamics:

system (fixed mass)



$$\delta q + \delta w = \delta u_e = \text{change in energy}$$

δq = heat added, δw = work done, δu_e = change in energy

- *adiabatic* process: no heat added or taken away
- *reversible* process: no dissipative phenomena occur
- *isentropic* process: adiabatic + reversible

for a reversible process

$$\delta w = -p dv$$

incremental change of volume due to boundary displacement

$$\Rightarrow \delta q - p dv = \delta u_e \quad (8.1)$$

(alternative form of 1st law)

Entropy and second law [section 8.1.3]

for an adiabatic process:

$$ds \geq 0 \quad (8.17)$$

First law: you cannot *win* Second law: you always *lose*

Calculation of entropy (for a reversible process)

$$\delta q = Tds \quad \Rightarrow \quad Tds - pdv = du_e$$

$$\Rightarrow \quad Tds = du_e + pdv \quad (8.18)$$

use enthalpy $h = u_e + pv$

$$\Rightarrow dh = du_e + pdv + vdp$$

$$\Rightarrow Tds = dh - vdp$$

The above equation and 8.18 are alternative forms of 1st law

for a thermally perfect gas: $dh = c_p dT$ (8.10)

$$\Rightarrow ds = c_p \frac{dT}{T} - \frac{vdp}{T}$$

assume:

$pv = RT$ (ideal gas law)

$$\Rightarrow ds = c_p \frac{dT}{T} - R \frac{dp}{p} \Rightarrow s_2 - s_1 = \int_{T_1}^{T_2} c_p \frac{dT}{T} - R \ln \frac{p_2}{p_1}$$

for a calorically perfect gas ($c_p = \text{constant}$)

$$\Rightarrow s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \quad (\text{equivalent to 8.19b})$$

similarly: *(use $p\nu = RT$, $c_p = \frac{\gamma R}{\gamma - 1}$; $c_v = \frac{R}{\gamma - 1}$ 8.16)*

$$\Rightarrow s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{\nu_2}{\nu_1} \quad \nu = 1/\rho \quad (\text{equivalent to 8.19c})$$

useful eqs to evaluate entropy increases

Isentropic relations

put $s_1 = s_2$ in (8.19a)

$$\Rightarrow 0 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \Rightarrow \ln \frac{p_2}{p_1} = \frac{c_p}{R} \ln \frac{T_2}{T_1}$$

$$\Rightarrow \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{c_p/R}$$

from (8.16) $\frac{c_p}{R} = \frac{\gamma}{\gamma - 1}$

$$\Rightarrow \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}$$

similarly:

$$\Rightarrow \frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1} \right)^{1/(\gamma-1)}$$

in summary:

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1} \right)^\gamma = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}$$

isentropic
relations

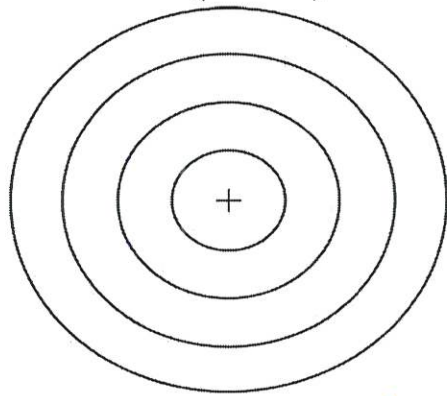
or:

(8.20)

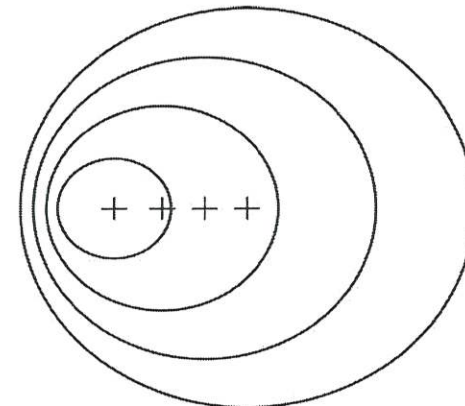
$$\frac{p}{\rho^\gamma} = \text{constant}; \quad \frac{T^{\gamma/(\gamma-1)}}{p} = \text{constant}; \quad \frac{T}{\rho^{(\gamma-1)}} = T v^{(\gamma-1)} = \text{constant}$$

(energy relation for an isentropic process) many processes can be assumed isentropic.

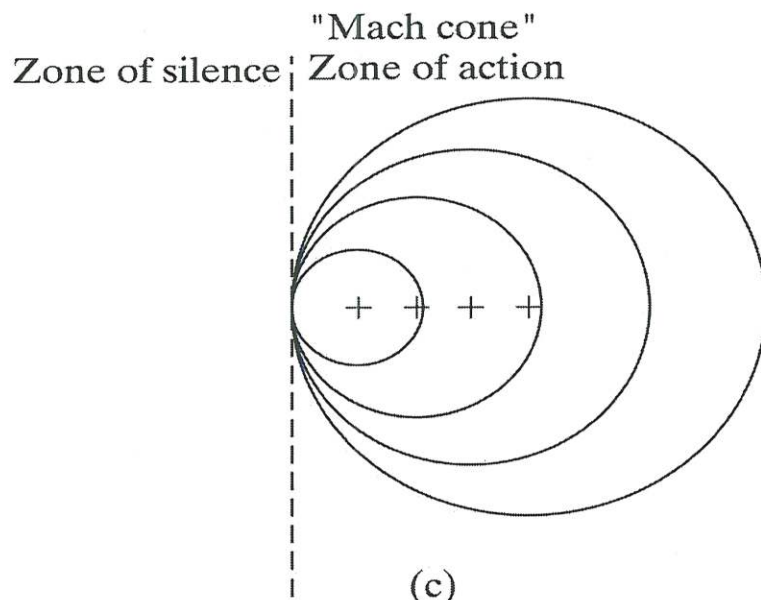
Speed of sound (8.1.4)



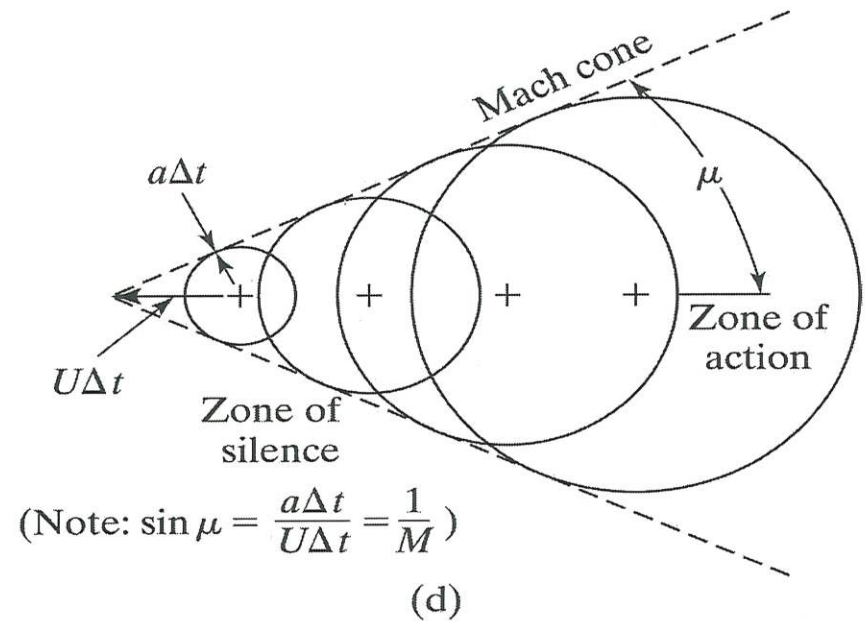
(a)

 $M = 0$ 

(b)

 $M < 1$ 

(c)

 $M = 1$ 

(d)

 $M > 1$

(a) no external velocity

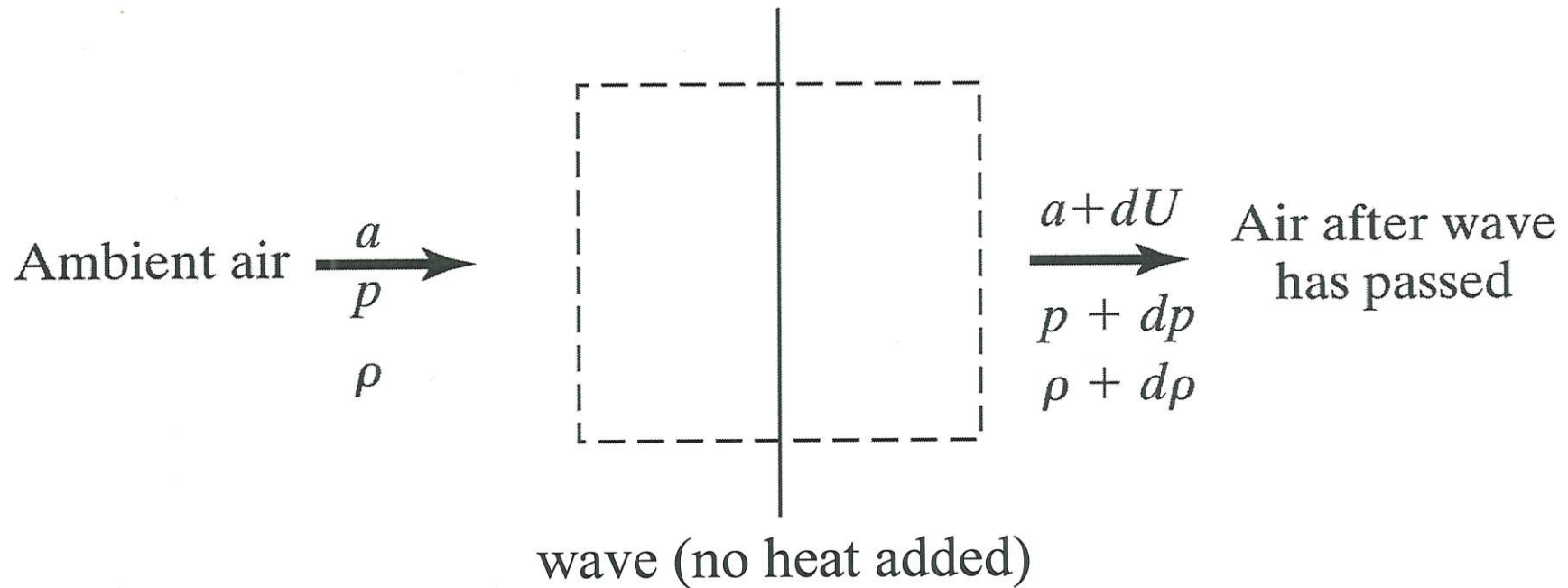
(b) Velocity $U < a$ - subsonic speed source stays inside family of circular sound waves ($Ut < at$)

(d) Velocity $U > a$ - supersonic speed source outside family of circles wave fronts form a disturbance envelope

$\Rightarrow$ Mach wave/cone

Mach angle:

$$\sin \mu = \frac{at}{Ut} = \frac{a}{U} = \frac{1}{M} \quad \Rightarrow \quad \mu = \sin^{-1} \left(\frac{1}{M} \right) \quad (8.22)$$



assume a weak wave with speed a ; changes to speed, pressure density, temperature are small $\Rightarrow$ isentropic

continuity: $\rho a = (\rho + d\rho)(a + dU)$ (8.23)

momentum: $p + \rho a^2 = (p + dp) + (\rho + d\rho)(a + dU)^2$ (8.24)

ignoring 2nd order terms, after some algebra

$$\Rightarrow a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} \text{ - isentropic} \quad (8.26)$$

for a calorically perfect gas: $p = \rho^\gamma \times \text{constant}$

$$\Rightarrow a = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\gamma R T} \quad (8.28)$$

for standard sea level: $a_s = 340.9 \text{ m/s} = 1117 \text{ ft/s}$

$$M \equiv \frac{U}{a}$$

↓

Mach number

$M < 1$ subsonic

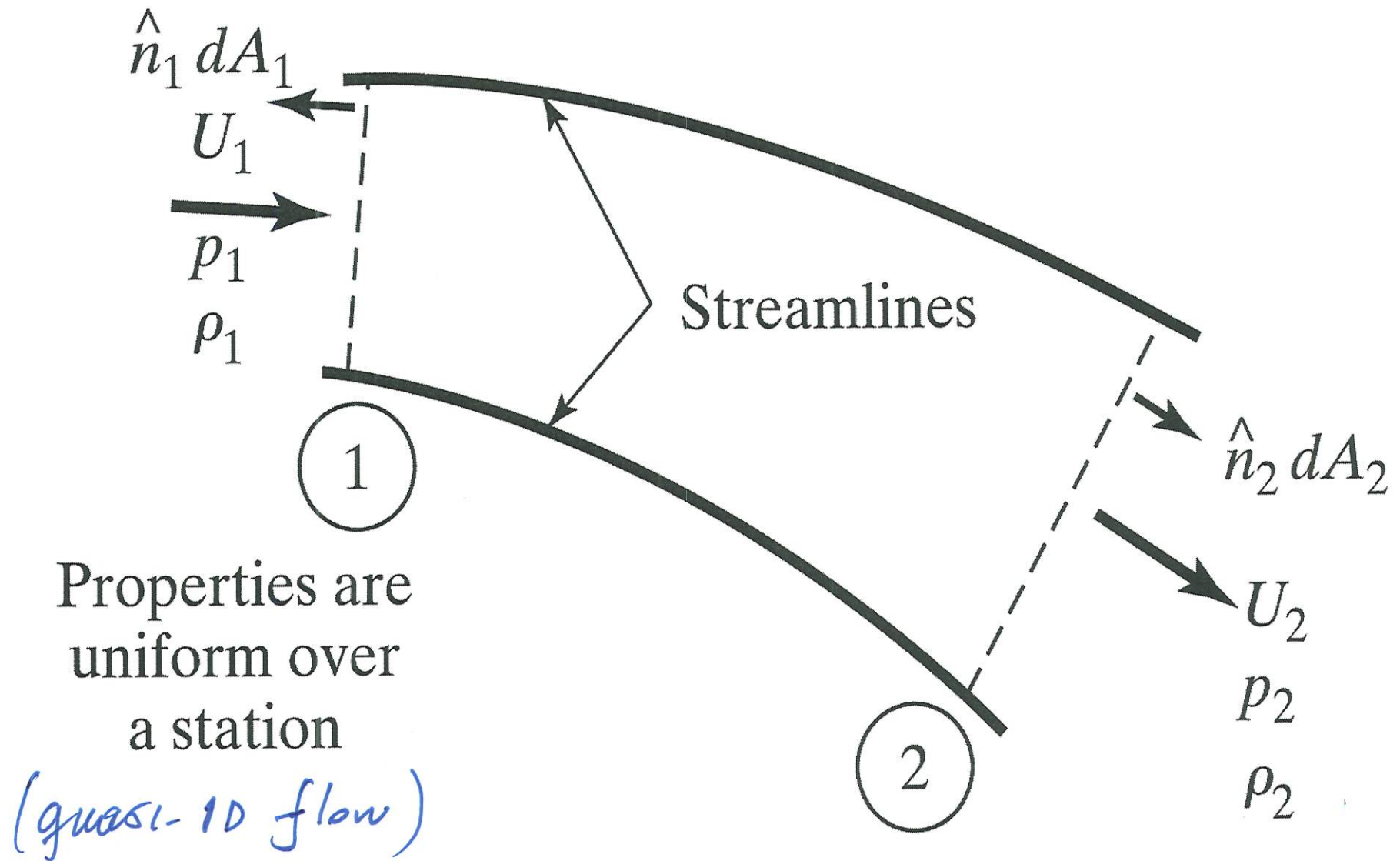
$M = 1$ sonic

$M > 1$ supersonic

for incompressible flow:

$$M = 0 \quad ; \quad a \rightarrow \infty$$

Adiabatic Flow in a variable Area Streamtube [8.2]



Energy equation:

$$p_1 U_1 A_1 + u_{e1} \rho_1 U_1 A_1 + \frac{U_1^2}{2} \rho_1 U_1 A_1 = p_2 U_2 A_2 + u_{e2} \rho_2 U_2 A_2 + \frac{U_2^2}{2} \rho_2 U_2 A_2$$

Using continuity

$$\rho_1 U_1 A_1 = \rho_2 U_2 A_2$$

and enthalpy definition

$$h = u_e + p v \quad (v = 1/\rho)$$

$$\Rightarrow H_t = h_1 + \frac{U_1^2}{2} = h_2 + \frac{U_2^2}{2} \quad (8.29)$$

$H_t \equiv$ total (stagnation) enthalpy

For a perfect gas:

$$c_p T_1 + \frac{U_1^2}{2} = c_p T_2 + \frac{U_2^2}{2} \quad (8.30)$$

$$T_1 + \frac{U_1^2}{2c_p} = T_2 + \frac{U_2^2}{2c_p} \quad (8.31)$$

$$T_t = T + \frac{U^2}{2c_p} = T_{static} + T_{dynamic} = \text{constant} \quad (8.32)$$

$$T_{t1} = T_{t2} \quad (8.33)$$

(adiabatic process)

From (8.32) using equation 8.16 for c_p

$$\frac{T_t}{T} = 1 + \frac{U^2}{2\gamma RT / (\gamma - 1)} = 1 + \frac{\gamma - 1}{2} M^2 \quad (8.34)$$

[Can be also derived starting momentum equation for isentropic inviscid flow-see text]

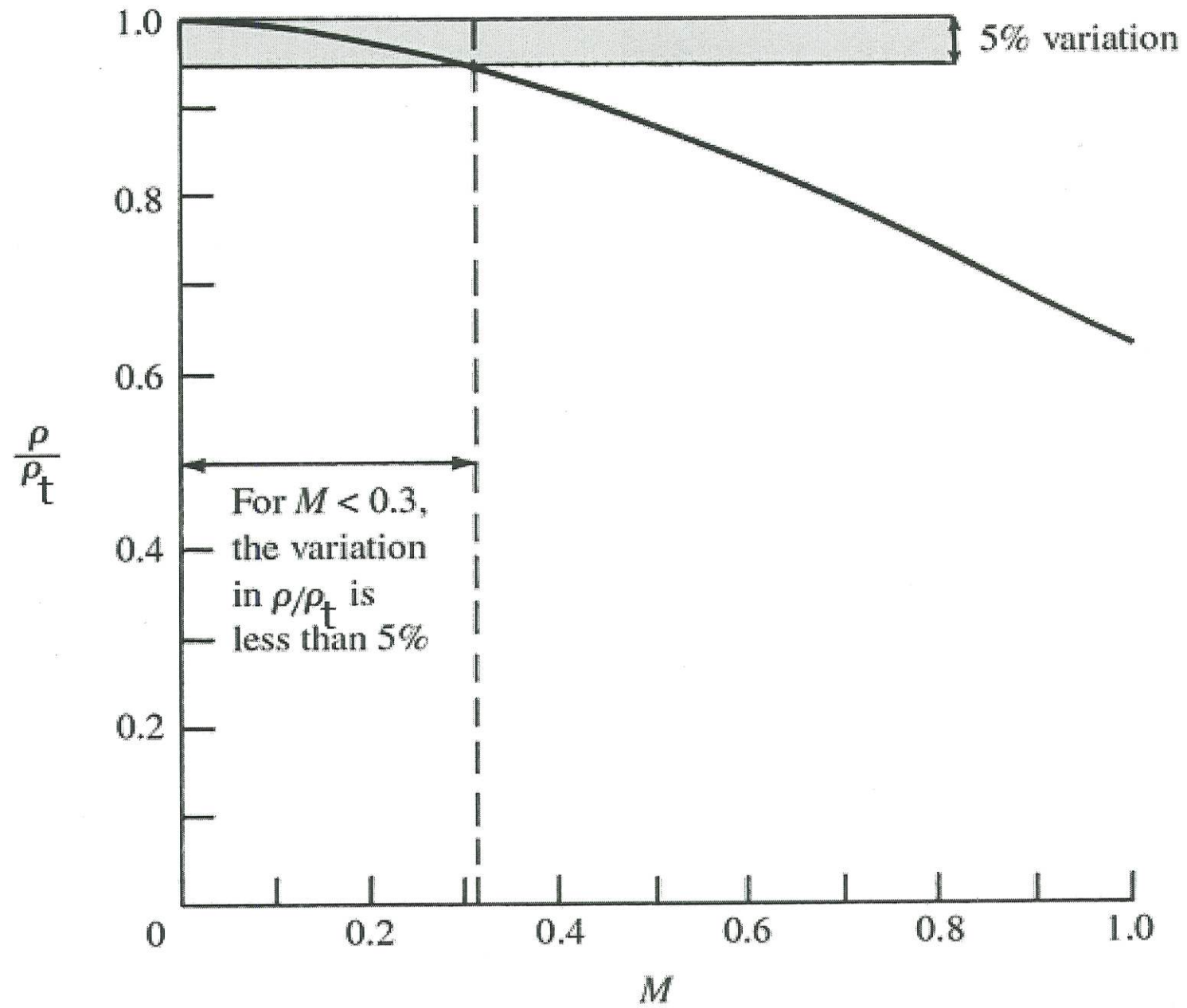
For isentropic flow: $\frac{p_t}{p} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma - 1)} \quad (8.36)$
(using 8.20)

$$\frac{\rho_t}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/(\gamma - 1)} \quad (8.37)$$

p_t and ρ_t are stagnation (total) pressure and density

from (8.36) 5% error in density for $M = 0.32$ *1% for $M = 0.14$*
10% for $M = 0.44$

8.34, 8.36, 8.37 are tabulated in Table 8.1 (isentropic tables)



binomial expansion on M^2 :

(for small M^2)

$$\frac{p_t}{p} = \left(1 + \frac{\gamma}{2} M^2 + \frac{\gamma}{8} M^4 + \dots \right) \quad (8.38)$$

for a low M :

$$\frac{p_t}{p} = \left(1 + \frac{\gamma}{2} M^2 \right) \quad (8.39)$$

for a perfect gas:

$$\frac{p_t}{p} = 1 + \frac{\gamma U^2}{2\gamma RT} = 1 + \frac{U^2}{2p/\rho} \quad (8.40)$$

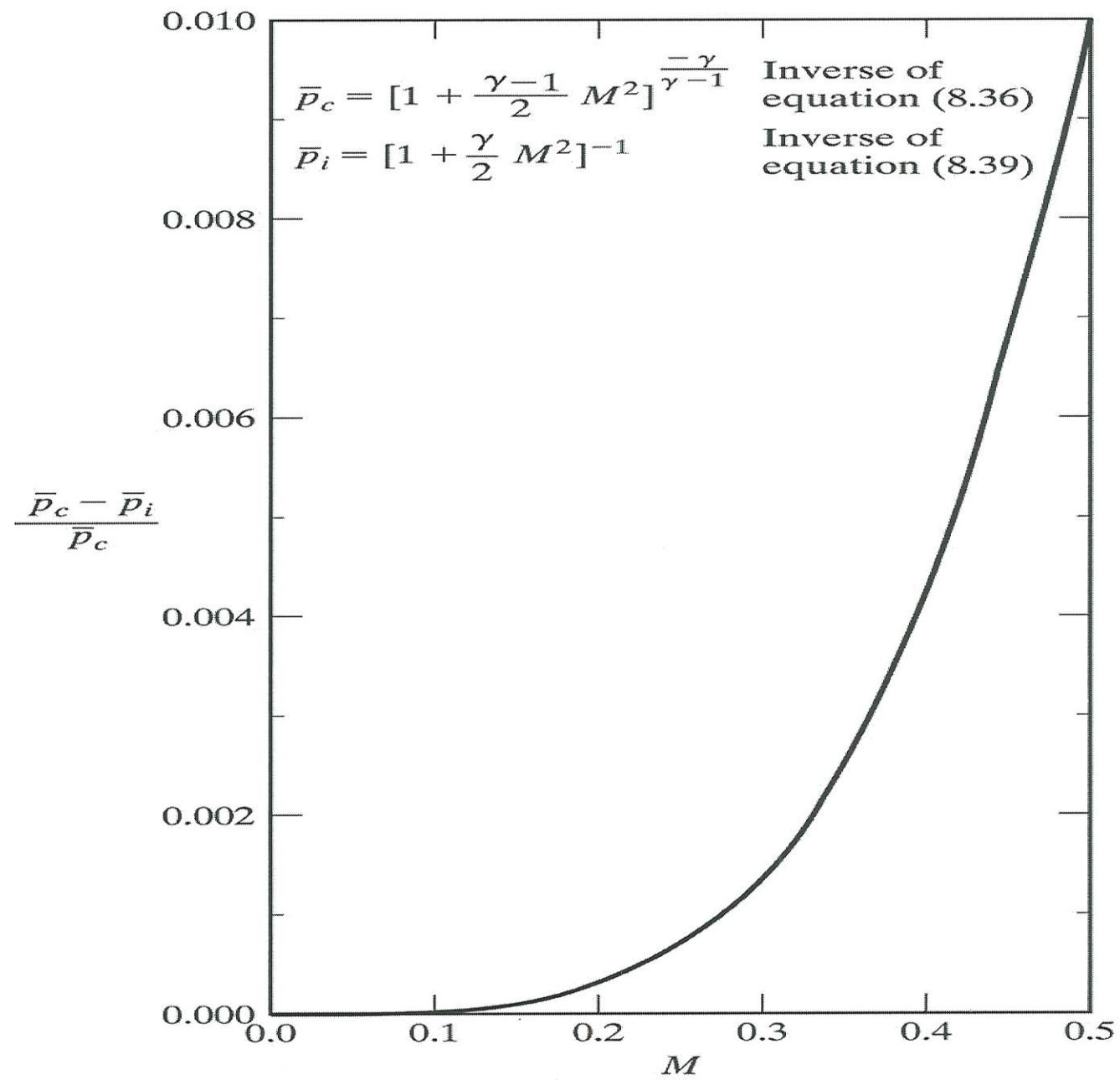
rearranging:

$$p_t = p + \frac{\rho U^2}{2} \left(+ \frac{\gamma p}{8} M^4 + \dots \right) \quad (8.41)$$

(Bernoulli's equation)

error $\approx 1\%$ for $M = 0.5$

0.15% for $M = 0.3$



Example: In the reservoir of a supersonic wind tunnel, the velocity is negligible, and the temperature is 1000K. The temperature at the nozzle exit is 600K. Assuming adiabatic flow through the nozzle, calculate the velocity at the exit.

$U=0$
(total conditions)

Reservoir:

$$H_t = C_p T_t$$

Exit:

$$(H_t)_{res} = (H_t)_{exit} = C_p T_e + \frac{V_e^2}{2}$$

Adiabatic flow:

$$(T_t)_{res} = (T_t)_{exit}$$

$$\Rightarrow C_p (T_t)_{res} = C_p T_e + \frac{V_e^2}{2}$$

$$\left(C_{p,air} = \frac{\gamma R}{\gamma - 1} = 1004.5 \frac{\text{Joules}}{\text{kg}^\circ\text{K}} \right)$$

$$\Rightarrow V_e = \sqrt{2 C_p ((T_t)_{res} - T_e)} = 896.4 \text{ m/s}$$